

AI & ML Internship – Task 3

Exploratory Data Analysis (EDA) using Tips Dataset

Introduction

Exploratory Data Analysis (EDA) is an important process in Data Science used to understand data through statistical summaries and visualizations. It helps identify patterns, trends, correlations, and anomalies before building Machine Learning models.

In this task, the **Tips Dataset** was analyzed using various visualization techniques such as Histograms, Boxplots, Scatterplots, Heatmaps, and Pairplots to extract meaningful insights from the data.

Objectives

- Understand dataset structure
- Perform Univariate Analysis
- Perform Bivariate Analysis
- Visualize correlations using Heatmap
- Identify patterns and outliers
- Generate meaningful insights from dataset

Tools & Technologies Used

Tool	Purpose
Python	Programming Language
Pandas	Data Analysis
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Jupyter Notebook	Development Environment

Dataset Used

Tips Dataset

The Tips dataset contains restaurant bill information such as:

- Total bill
- Tip amount
- Gender
- Smoking status
- Day
- Time
- Size of group

This dataset is commonly used for learning EDA and visualization concepts.

Step-by-Step Process

Step 1: Import Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load Dataset

```
df = sns.load_dataset('tips')
print(df.head())
```

Output

Displays first 5 rows of dataset.

Step 3: Dataset Information

```
print(df.info())
```

Purpose

Shows:

- Number of rows and columns
- Data types
- Missing values

Step 4: Statistical Summary

```
print(df.describe())
```

Output

Displays:

- Mean
- Standard deviation
- Minimum and maximum values

Step 5: Univariate Analysis

Histogram

```
plt.figure(figsize=(8,5))  
  
plt.hist(df['total_bill'], bins=20)  
  
plt.title("Distribution of Total Bill")  
plt.xlabel("Total Bill")  
plt.ylabel("Frequency")  
  
plt.show()
```

Purpose

Shows distribution of total bill values.

Boxplot

```
plt.figure(figsize=(8,5))  
sns.boxplot(x=df['total_bill'])  
plt.title("Boxplot of Total Bill")  
plt.show()
```

Purpose

Identifies outliers visually.

Step 6: Bivariate Analysis

Scatter Plot

```
plt.figure(figsize=(8,5))  
sns.scatterplot(x='total_bill',  
                y='tip',  
                hue='sex',  
                data=df)  
plt.title("Total Bill vs Tip")  
plt.show()
```

Purpose

Shows relationship between total bill and tip amount.

Step 7: Bar Plot

```
plt.figure(figsize=(8,5))  
sns.barplot(x='day',  
            y='total_bill',  
            data=df)  
plt.title("Average Total Bill by Day")  
plt.show()
```

Purpose

Compares average bill amounts across different days.

Step 8: Correlation Heatmap

```
plt.figure(figsize=(8,6))  
  
sns.heatmap(df.corr(numeric_only=True),  
            annot=True,  
            cmap='coolwarm')  
  
plt.title("Correlation Heatmap")  
plt.show()
```

Purpose

Displays correlation between numerical variables.

Step 9: Pairplot

```
sns.pairplot(df, hue='sex')  
plt.show()
```

Purpose

Visualizes relationships among numerical columns.

Step 10: Key Insights

Observations

- Higher total bills generally result in higher tips
 - Most customers paid medium-range bills
 - Few outliers are present in total bill values
 - Weekend days show higher average bills
 - Strong positive correlation exists between total bill and tip
-

Final Complete Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = sns.load_dataset('tips')

# Display dataset
print(df.head())

# Dataset information
print(df.info())

# Statistical summary
print(df.describe())

# Histogram
plt.figure(figsize=(8,5))

plt.hist(df['total_bill'], bins=20)

plt.title("Distribution of Total Bill")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")

plt.show()

# Boxplot
plt.figure(figsize=(8,5))

sns.boxplot(x=df['total_bill'])

plt.title("Boxplot of Total Bill")

plt.show()

# Scatter Plot
plt.figure(figsize=(8,5))

sns.scatterplot(x='total_bill',
                y='tip',
                hue='sex',
                data=df)

plt.title("Total Bill vs Tip")

plt.show()

# Bar Plot
plt.figure(figsize=(8,5))

sns.barplot(x='day',
            y='total_bill',
            data=df)

plt.title("Average Total Bill by Day")
```

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```
plt.show()

# Heatmap
plt.figure(figsize=(8,6))

sns.heatmap(df.corr(numeric_only=True),
             annot=True,
             cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()

# Pairplot
sns.pairplot(df, hue='sex')

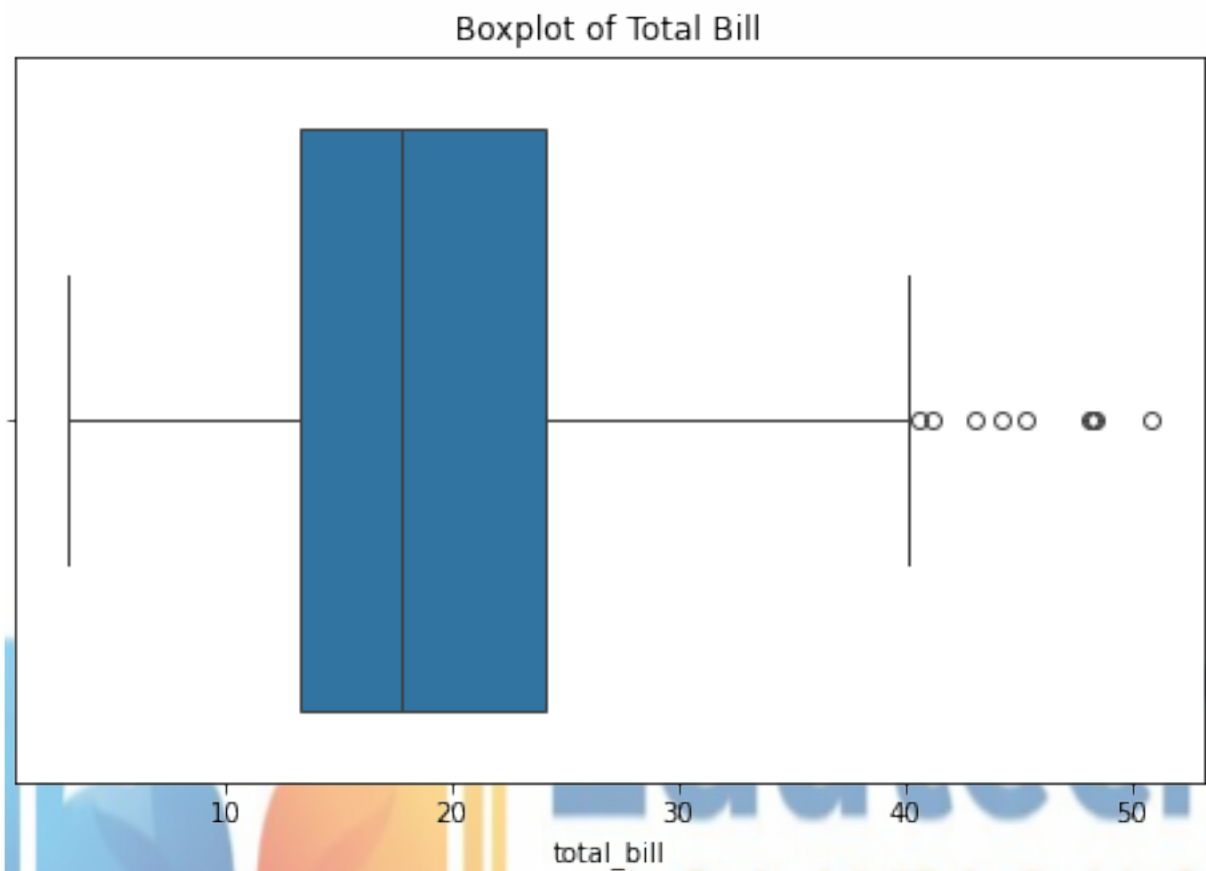
plt.show()

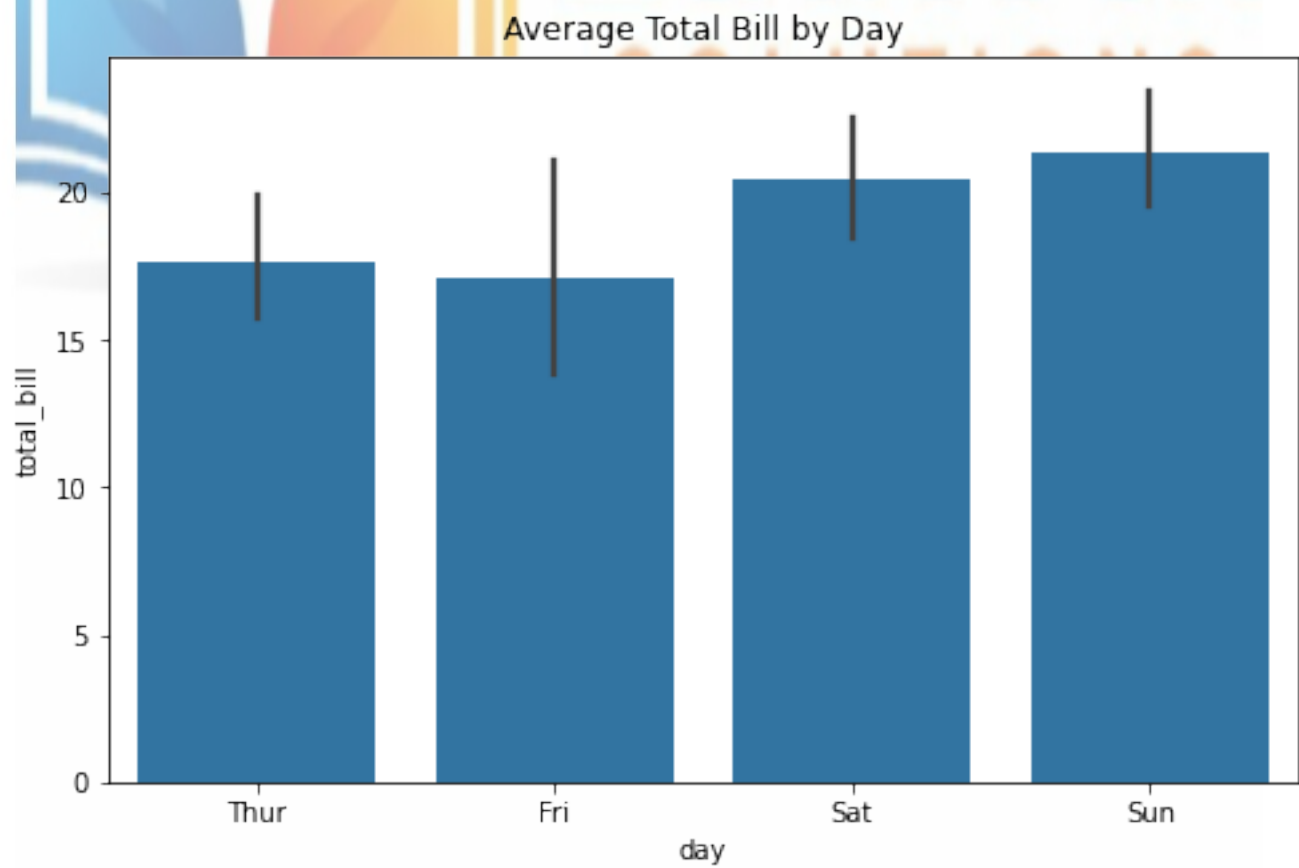
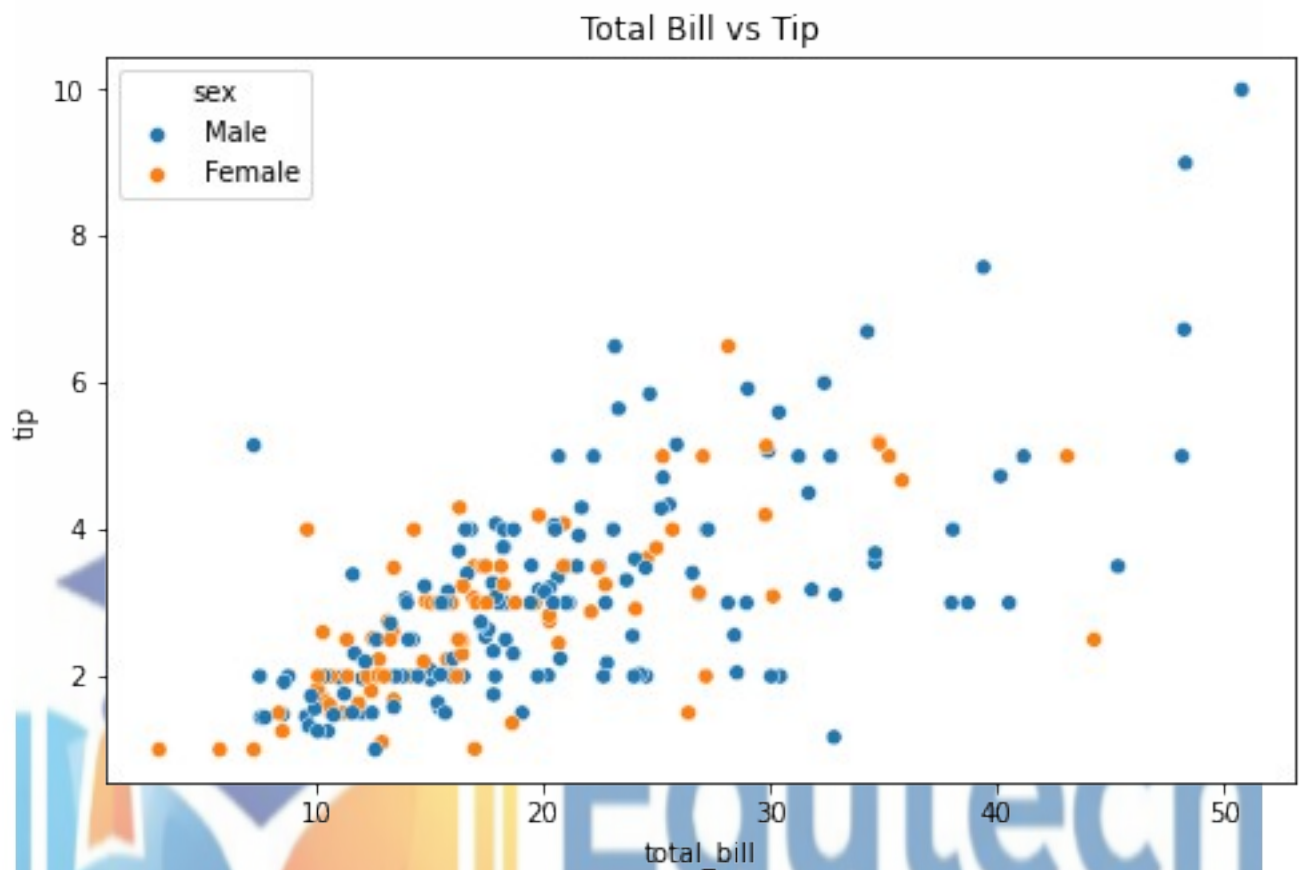
print("EDA Completed Successfully")
```

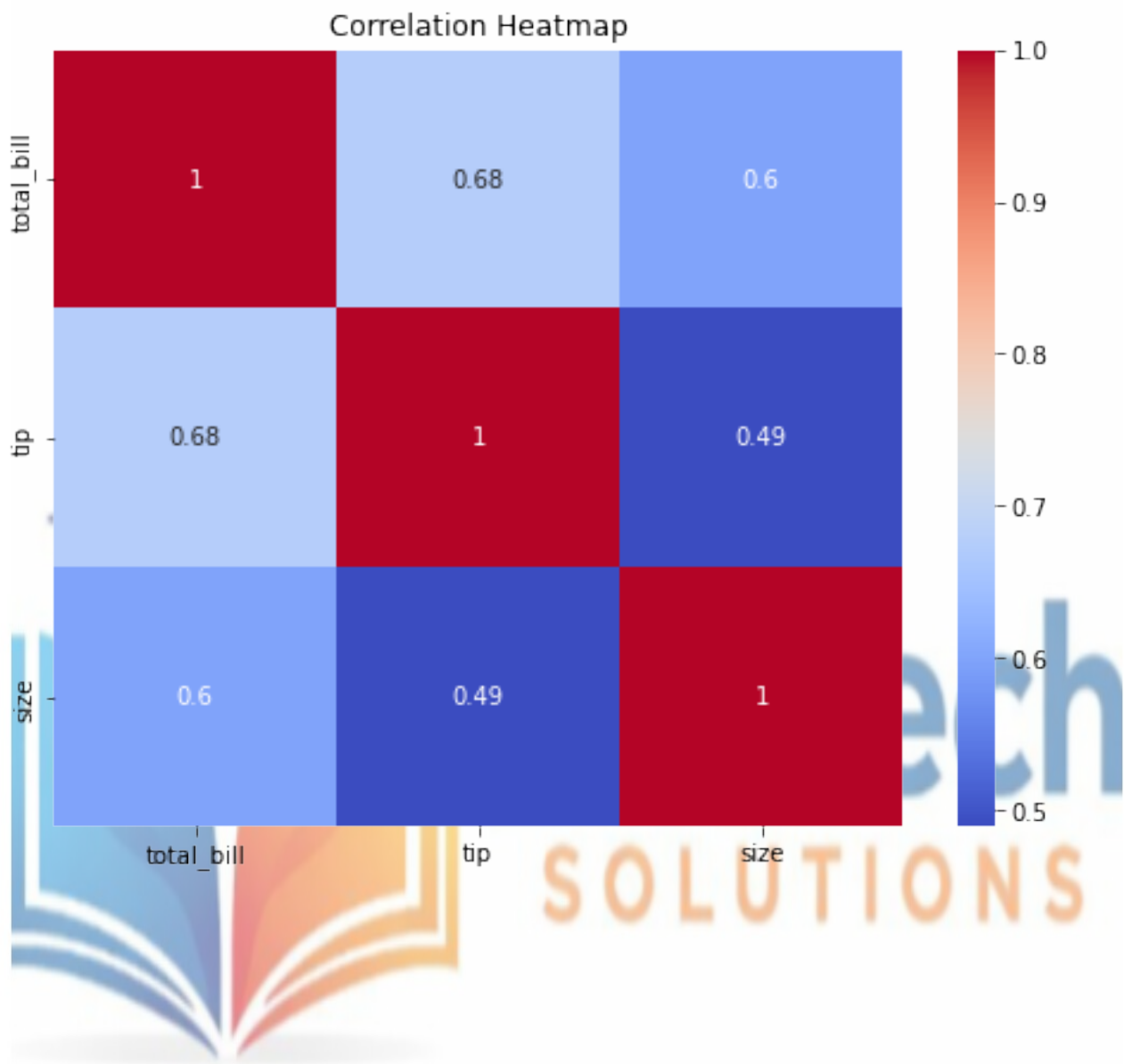
Expected Outputs

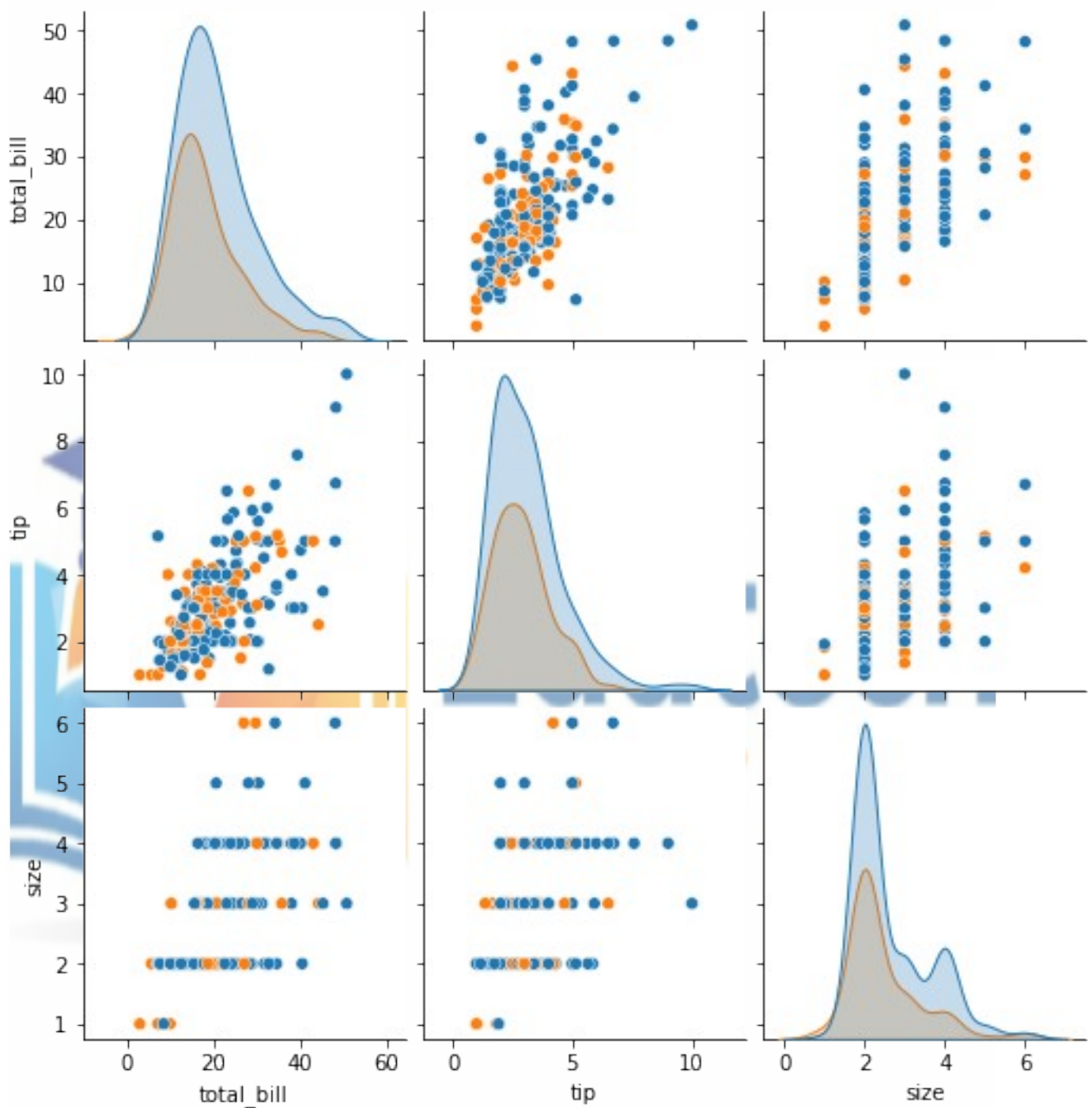
Histogram generated
Boxplot generated
Scatterplot generated
Barplot generated
Correlation Heatmap displayed
Pairplot displayed
Data insights identified successfully

Actual Outputs:









EDA Completed Successfully

[]:

Interview Questions & Answers – Task 3 (EDA)

1. What is Exploratory Data Analysis (EDA)?

Answer:

Exploratory Data Analysis (EDA) is the process of analyzing and visualizing datasets to understand patterns, trends, relationships, and anomalies before building Machine Learning models.

2. What is Univariate Analysis?

Answer:

Univariate Analysis analyzes a single variable independently.

Examples:

- Histogram
- Boxplot
- Countplot

Purpose:

To understand data distribution and detect outliers.

3. What is Bivariate Analysis?

Answer:

Bivariate Analysis studies the relationship between two variables.

Examples:

- Scatterplot

- Barplot
- Lineplot

Purpose:

To identify correlations and trends between variables.

4. What is Multivariate Analysis?

Answer:

Multivariate Analysis analyzes relationships among multiple variables simultaneously.

Example:

- Pairplot
 - Correlation Heatmap
-

5. What is the purpose of a Histogram?

Answer:

A Histogram shows the frequency distribution of numerical data.

It helps to:

- Understand data distribution
 - Detect skewness
 - Identify patterns
-

6. What is a Boxplot?

Answer:

A Boxplot is used to visualize:

- Median
- Quartiles

- Outliers

Purpose:

To detect extreme values in the dataset.

7. How do you identify outliers visually?

Answer:

Outliers can be identified using:

- Boxplots
- Scatterplots

Values far away from other observations are considered outliers.

8. What is a Correlation Heatmap?

Answer:

A Correlation Heatmap visually represents relationships between numerical variables using color intensity.

Correlation Values:

Value	Meaning
+1	Strong positive correlation
0	No correlation
-1	Strong negative correlation

9. What is the purpose of Pairplot?

Answer:

Pairplot visualizes relationships between all numerical variables in a dataset.

It helps:

- Compare features

- Detect clusters
 - Identify trends
-

10. Why is EDA important before modeling?

Answer:

EDA is important because it:

- Improves understanding of data
 - Detects missing values and outliers
 - Identifies patterns and correlations
 - Helps select suitable ML algorithms
 - Improves model accuracy
-

Additional Questions

11. Which libraries are used for EDA?

Answer:

- Pandas
 - Matplotlib
 - Seaborn
 - NumPy
-

12. What insights did you get from the Tips Dataset?

Answer:

- Higher bills generally lead to higher tips
- Weekend bills are usually higher

- Some outliers exist in total bill amounts
- Positive correlation exists between total bill and tip

Task Summary

In this task, Exploratory Data Analysis was performed on the Tips dataset using Python libraries such as Pandas, Matplotlib, and Seaborn. Various visualizations including Histograms, Boxplots, Scatterplots, Heatmaps, and Pairplots were generated to understand data distribution, relationships, and correlations.

The analysis helped identify patterns, trends, and outliers effectively, making the dataset ready for Machine Learning and further analysis.

Final Outcome

Successfully performed EDA on Tips Dataset
Extracted meaningful insights from data
Visualized patterns and relationships
Understood correlations and outliers
Prepared dataset for Machine Learning

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